Strategy for Trigonometric Equations

Exam code: 9MA0

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Strategy for trigonometric equations

What strategies can I use for solving trigonometric equations?

- You can solve trig equations in a variety of different ways
  - **Sketching a graph** (see **Graphs of Trigonometric Functions**)
  - Using **trigonometric identities** (see **Trigonometry – Simple Identities**)
  - Using the **CAST** diagram (see **Linear Trigonometric Equations**)
  - Factorising **quadratic** trig equations (see **Quadratic Trigonometric Equations**)
- You may be asked to use **degrees or radians** to solve trigonometric equations
  - Make sure your calculator is in the **correct mode**
  - Remember common angles
    - 90° is ½π radians
    - 180° is π radians
    - 270° is 3π/2 radians
    - 360° is 2π radians
- If you're having trouble solving a trig equation, this flowchart might help:

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Help me understand

YES

NO

TRANSFORM THE RANGE

DON'T FORGET TO TRANSFORM YOUR SOLUTIONS BACK AT THE END

CAN YOU **REARRANGE** EVERYTHING OVER TO ONE SIDE AND **FACTORISE** (IF NEEDED) TO GET DOWN TO SIMPLE sin, cos OR tan FUNCTIONS?

YES

NO

DOES IT INVOLVE AN **IDENTITY**?

$\tan x \equiv \frac{\sin x}{\cos x}$ OR $\sin^2 x + \cos^2 x \equiv 1$

USE THESE TO CONVERT EVERYTHING INTO ONE TRIG FUNCTION

IS IT **LINEAR** OR **QUADRATIC**?

LINEAR

USE $\tan x \equiv \frac{\sin x}{\cos x}$

CAN I REARRANGE IT? IS THERE A COMMON FACTOR?

QUADRATIC

USE $\sin^2 x + \cos^2 x \equiv 1$

CAN I REARRANGE IT? DO I NEED TO FACTORISE? YOU SHOULD END UP WITH TWO SOLUTIONS

DO ALL MY SOLUTIONS ACTUALLY EXIST?

REMEMBER SOLUTIONS TO $\sin x = k$ AND $\cos x = k$ ONLY EXIST WHERE $-1 \leqslant k \leqslant 1$ WHEREAS SOLUTIONS TO $\tan x = k$ EXIST FOR **ALL VALUES** OF k

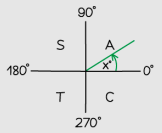
FIND THE **PRINCIPLE VALUE**

ARE SOLUTIONS WITHIN GIVEN RANGE? DO I HAVE **ALL** SOLUTIONS?

YES

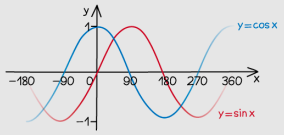
NO

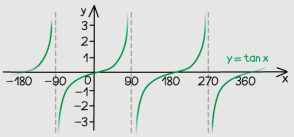
USE **CAST**



SKETCH THE GRAPH

USE YOUR KNOWLEDGE OF GRAPH SYMMETRIES AND PERIODICITY TO FIND ALL OTHER SOLUTIONS





CHECK WHETHER YOU HAVE FOUND **ALL SOLUTIONS** AND THAT THEY ARE **WITHIN THE GIVEN RANGE**
(DON'T FORGET TO TRANSFORM YOUR SOLUTIONS BACK AT THE END IF THEY WERE FROM A TRANSFORMED RANGE)

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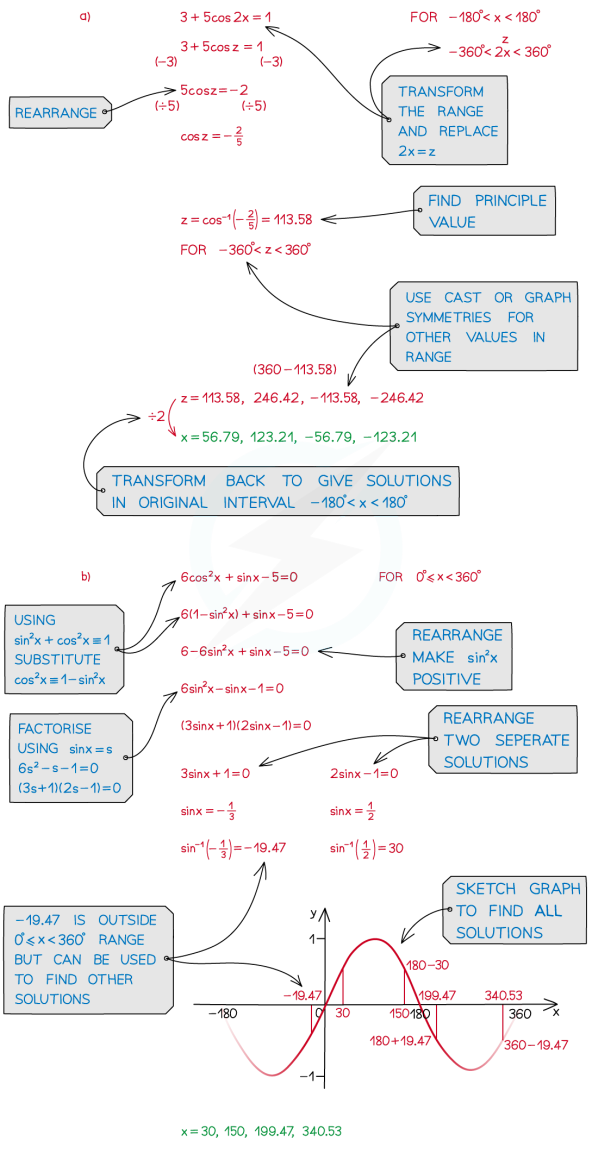
Examiner Tips and Tricks

- Don't forget to check the function range and ensure you have included all possible solutions.
- If the question involves a function of x or θ, make sure you transform the range first (and ensure you transform your solutions back again at the end!).

Worked Example

a) Solve $3 + 5\cos 2x = 1$, for $-180° < x < 180°$.

b) Solve $6\cos^2 x + \sin x - 5 = 0$, for $0 \leq x < 360°$.



x = 30, 150, 199.47, 340.53

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